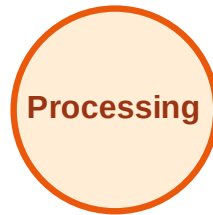


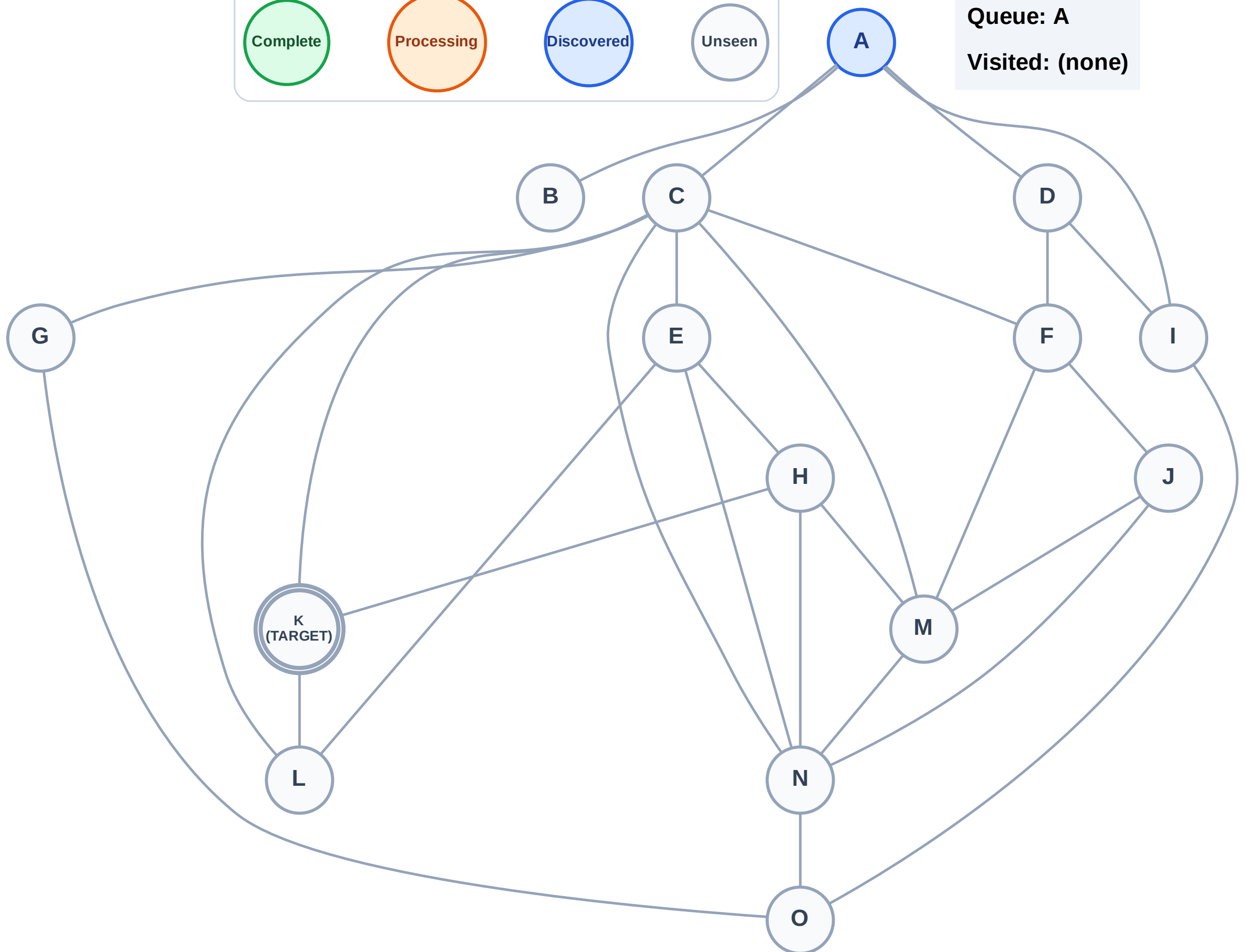
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



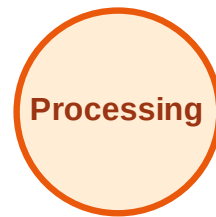
Queue: A
Visited: (none)



BFS Traversal

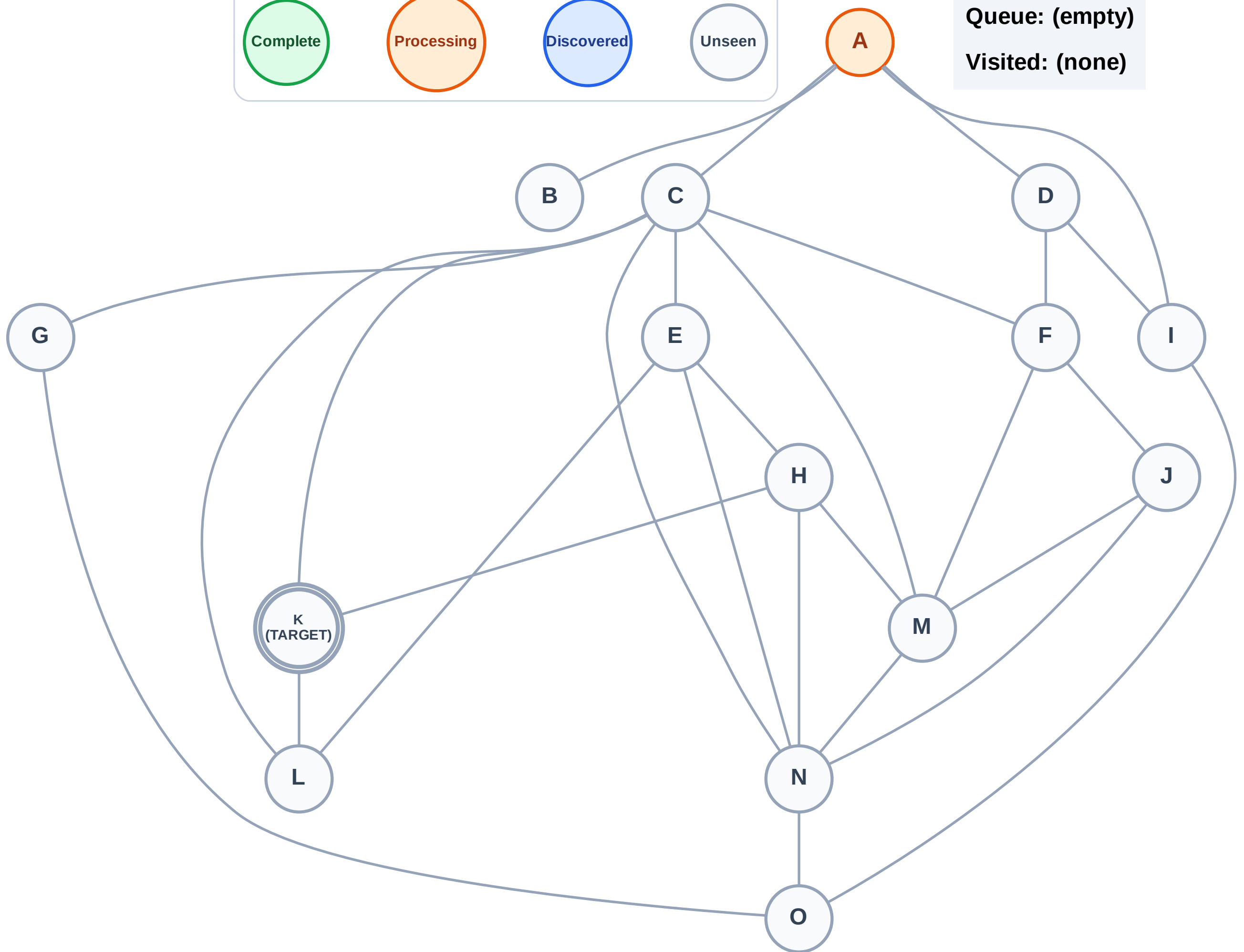
Step 2: Process node A

Legend



Queue: (empty)

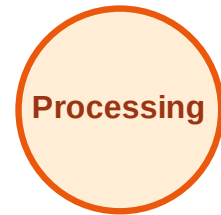
Visited: (none)



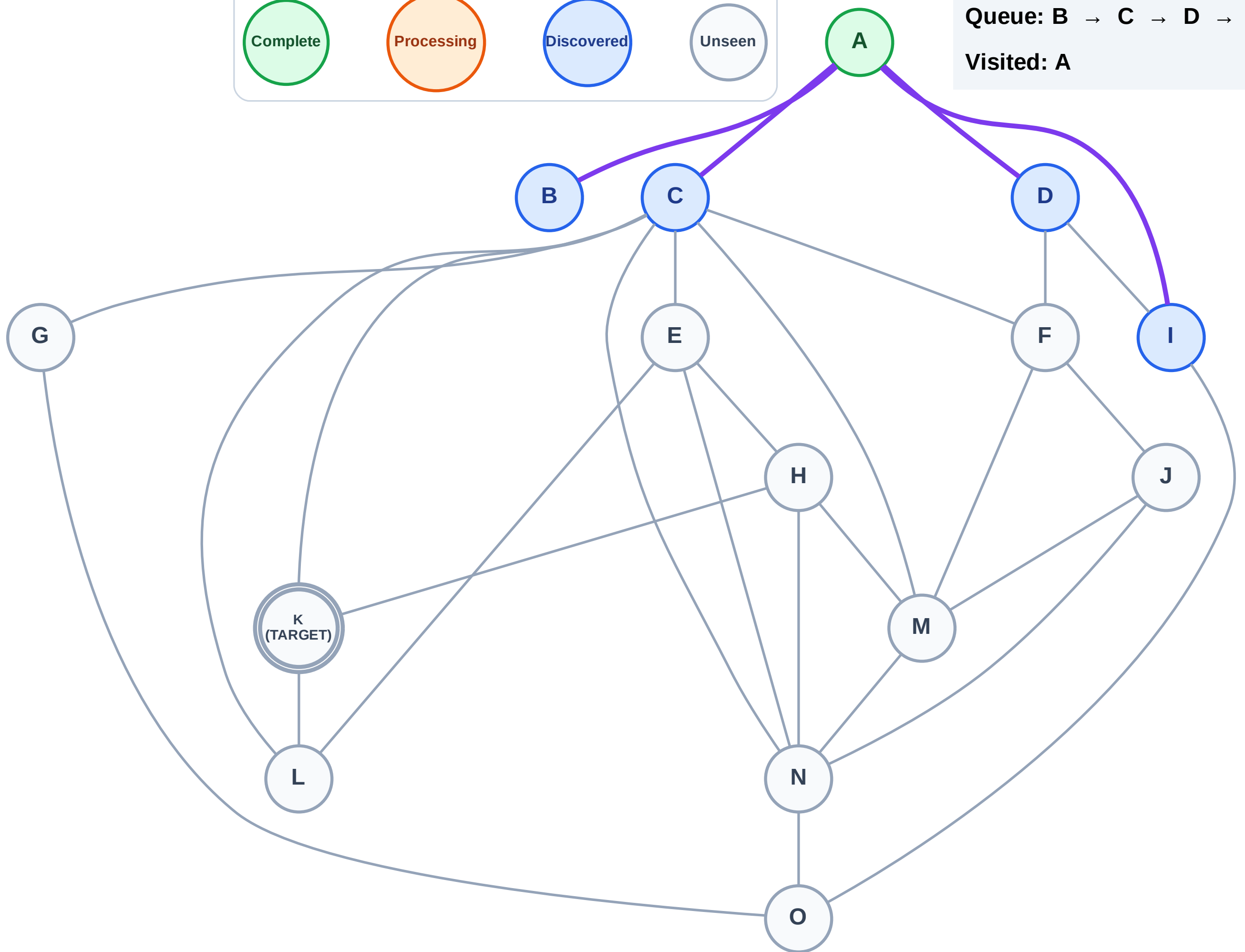
BFS Traversal

Step 3: Discover B, C, D, I from A

Legend

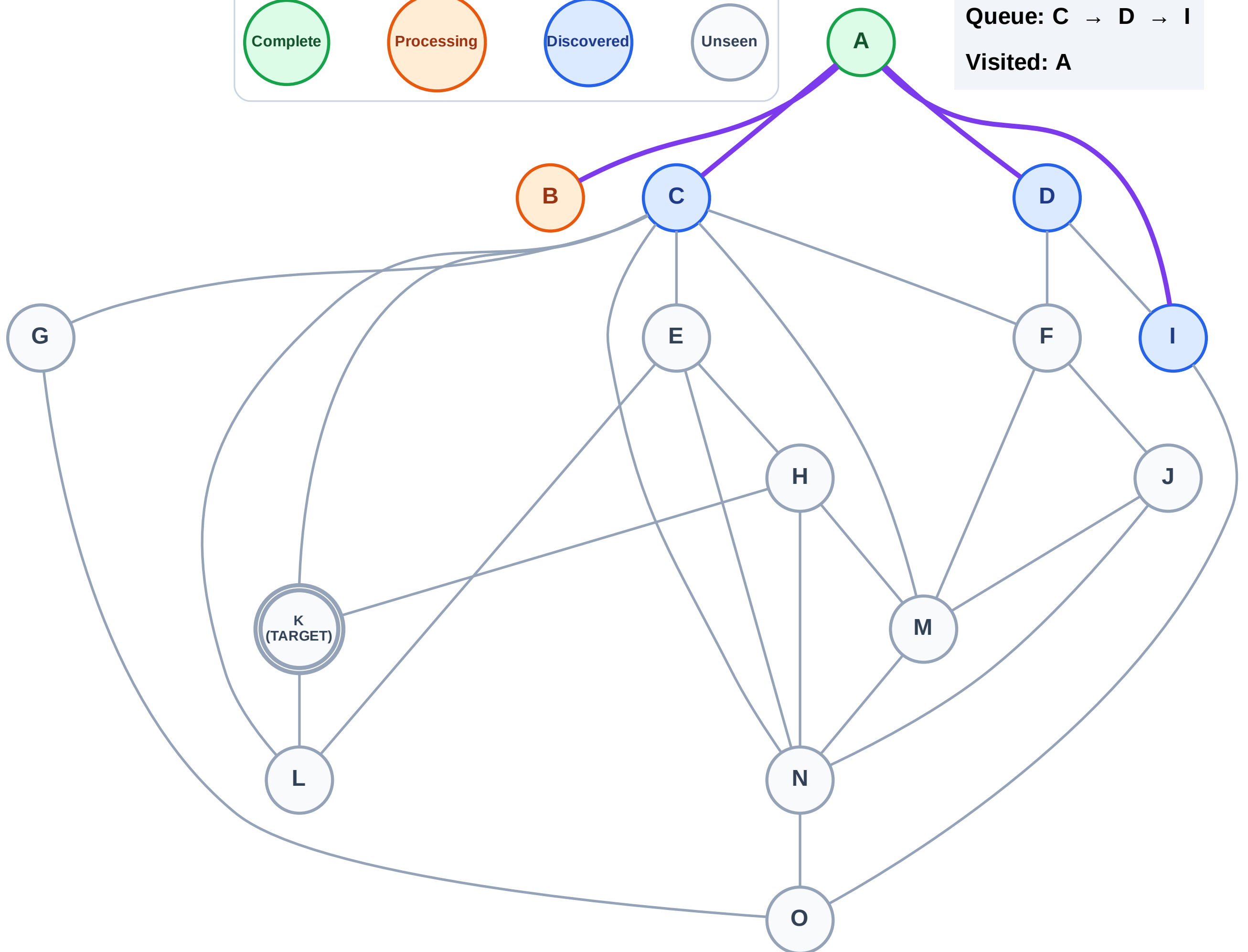
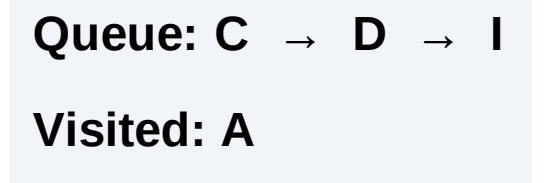


Queue: B → C → D → I
Visited: A



Step 4: Process node B

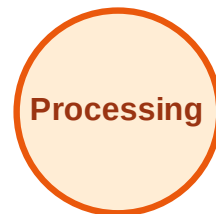
Legend



BFS Traversal

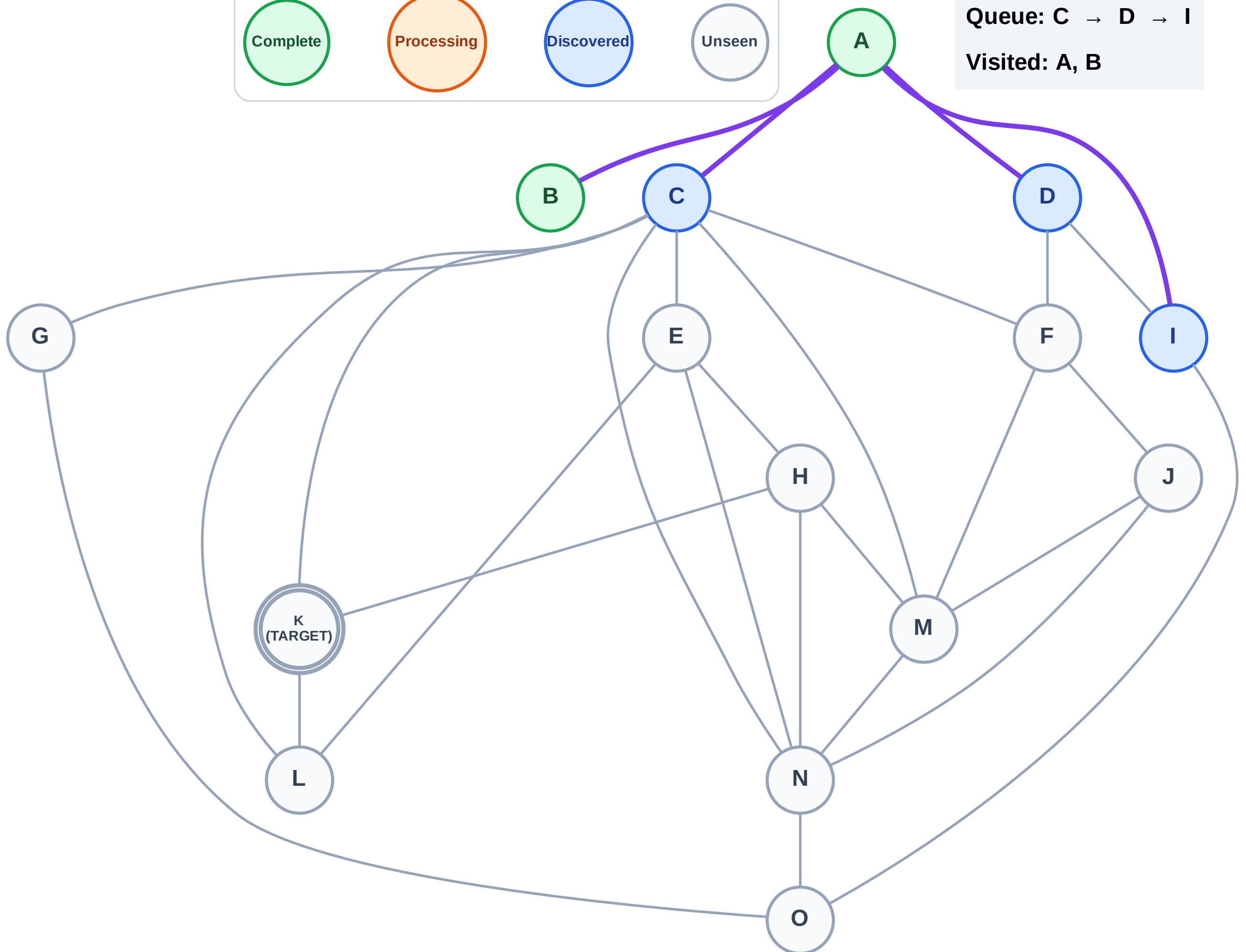
Step 5: No new neighbors from B

Legend



Queue: C → D → I

Visited: A, B



Step 6: Process node C

Step 6: Process node C

Legend

Complete

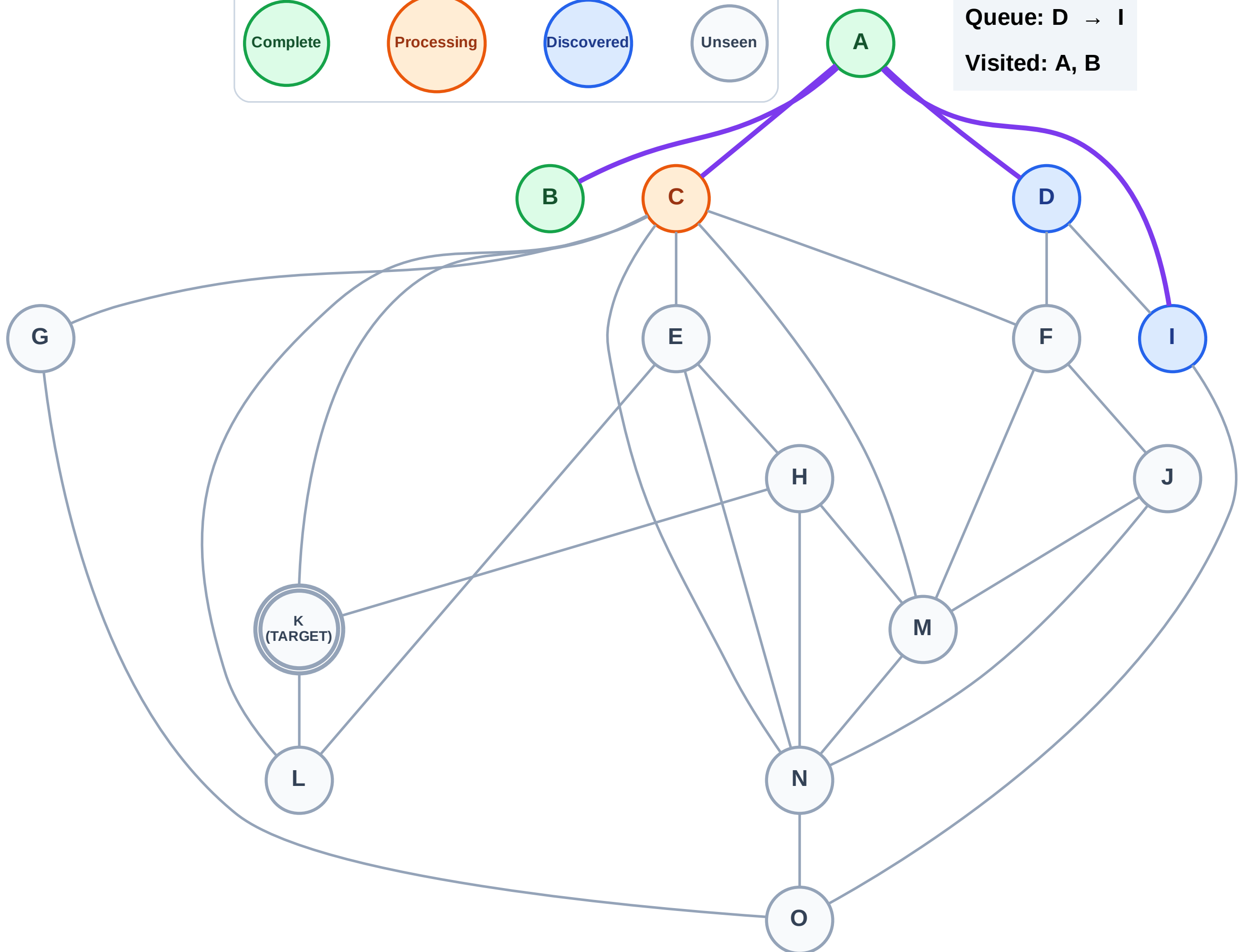
Processing

Discovered

Unseen

Queue: D → I

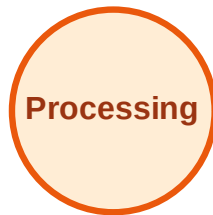
Visited: A, B



BFS Traversal

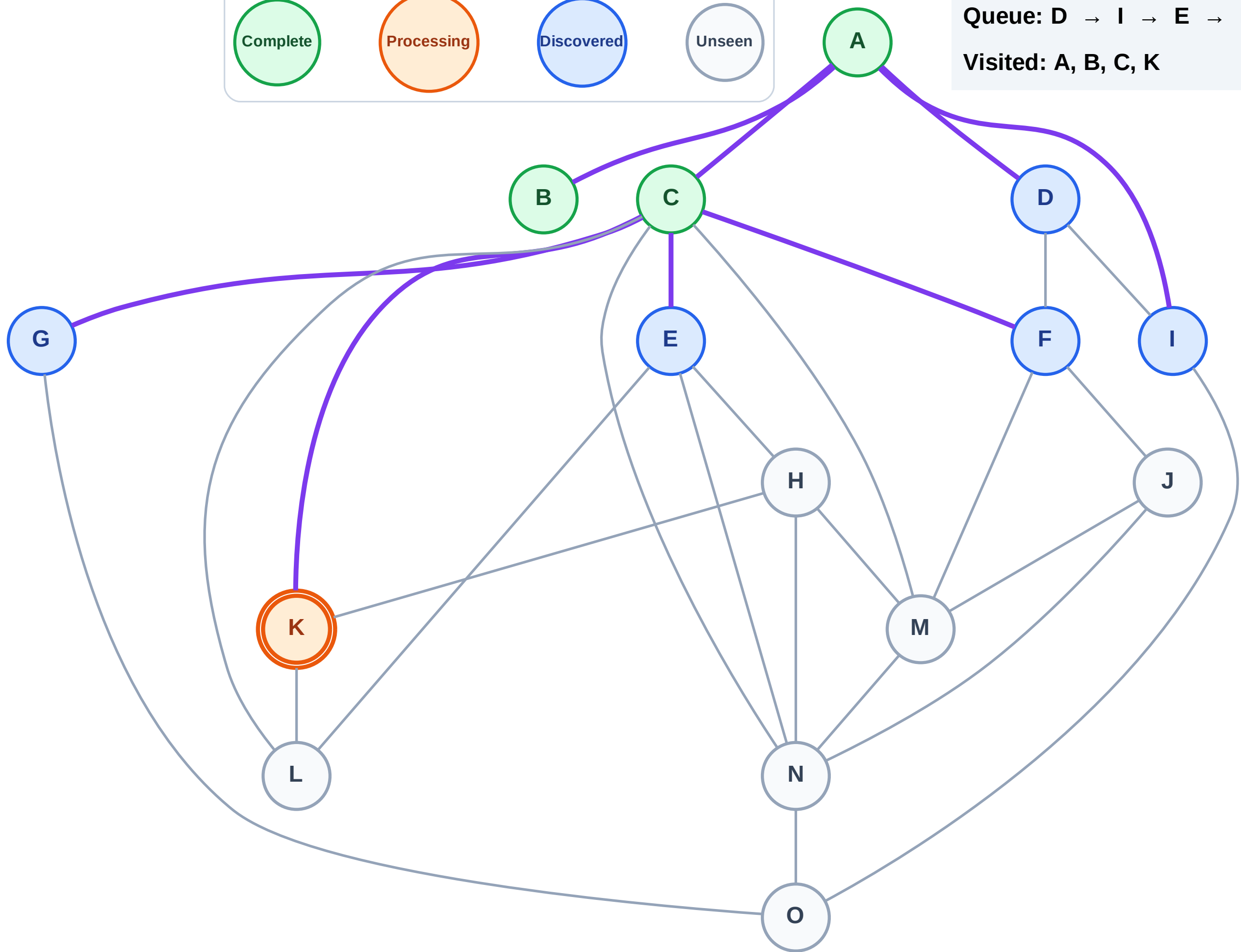
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Queue: D → I → E → F → G

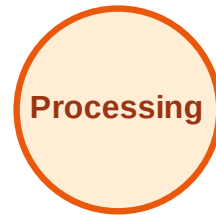
Visited: A, B, C, K



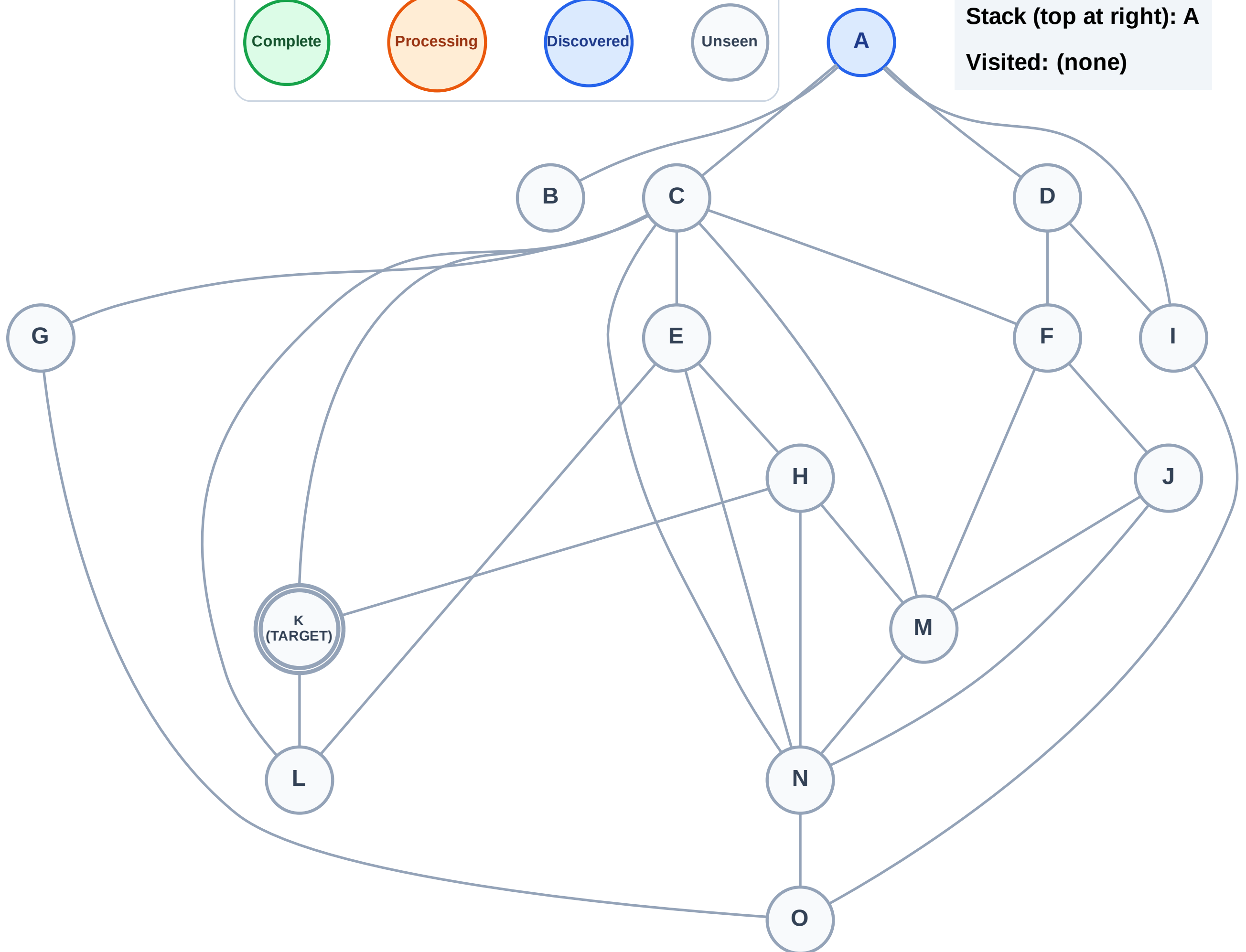
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



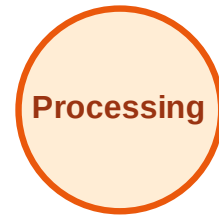
Stack (top at right): A
Visited: (none)



DFS Traversal

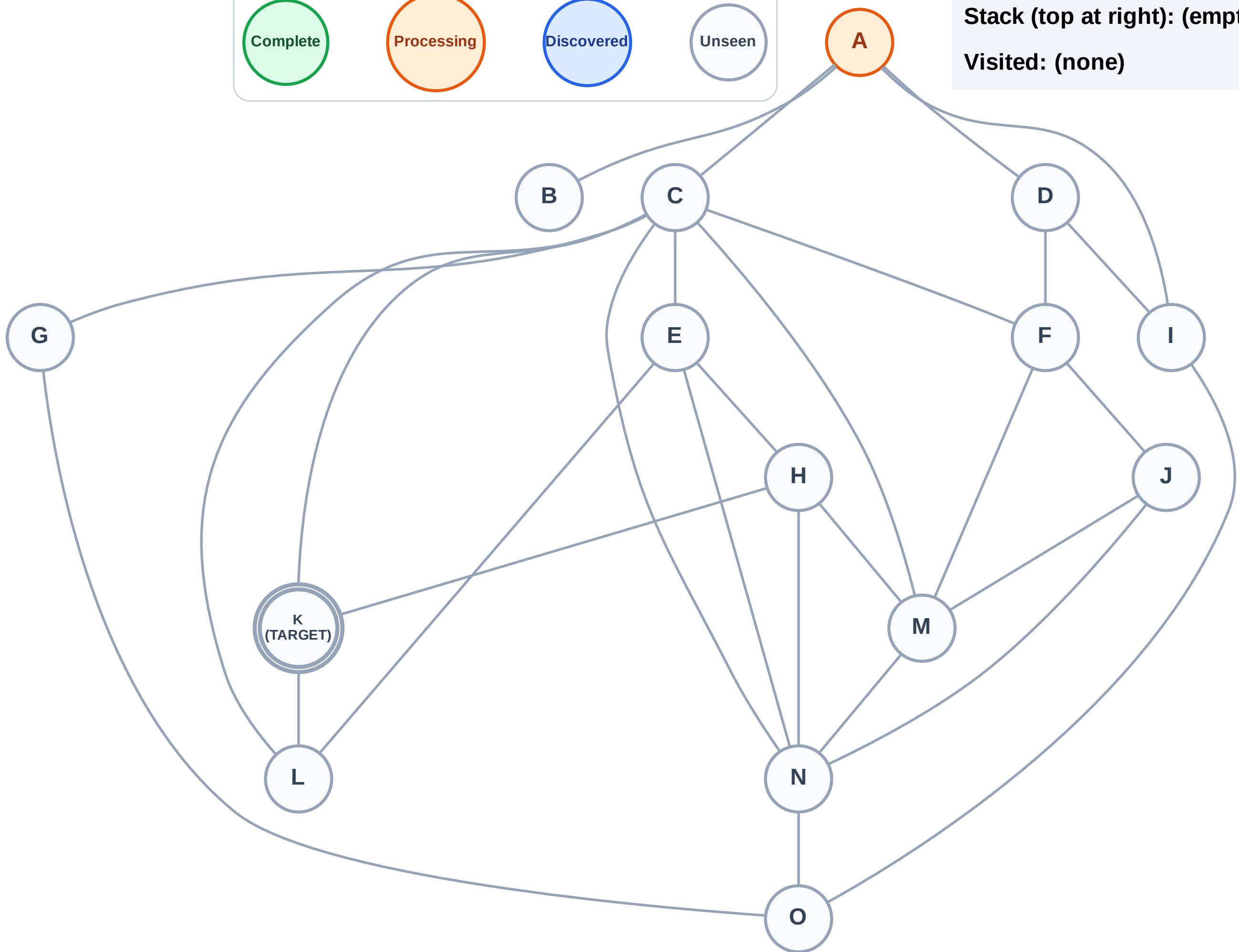
Step 2: Process node A

Legend



Stack (top at right): (empty)

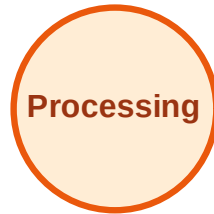
Visited: (none)



DFS Traversal

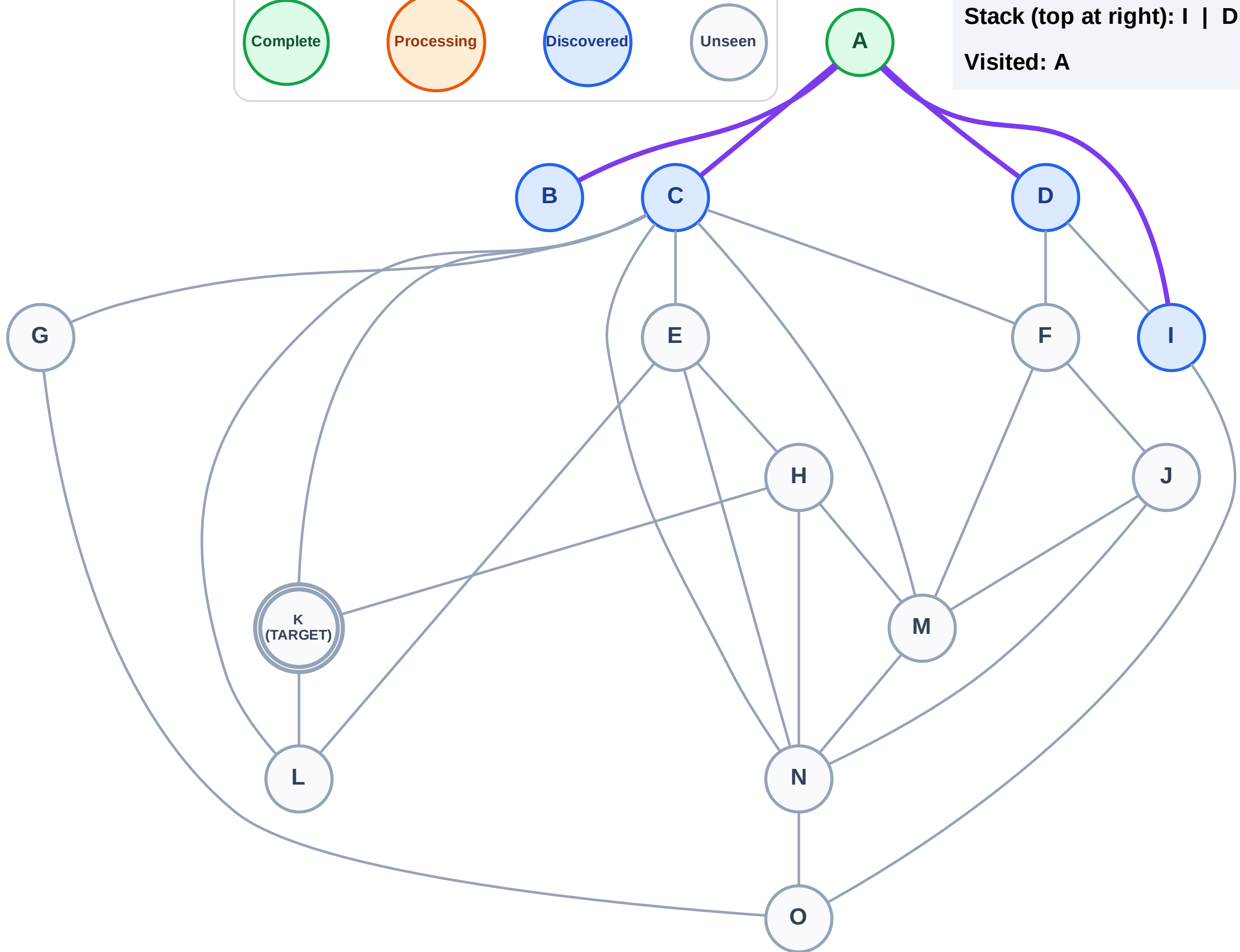
Step 3: Discover I, D, C, B from A

Legend



Stack (top at right): I | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

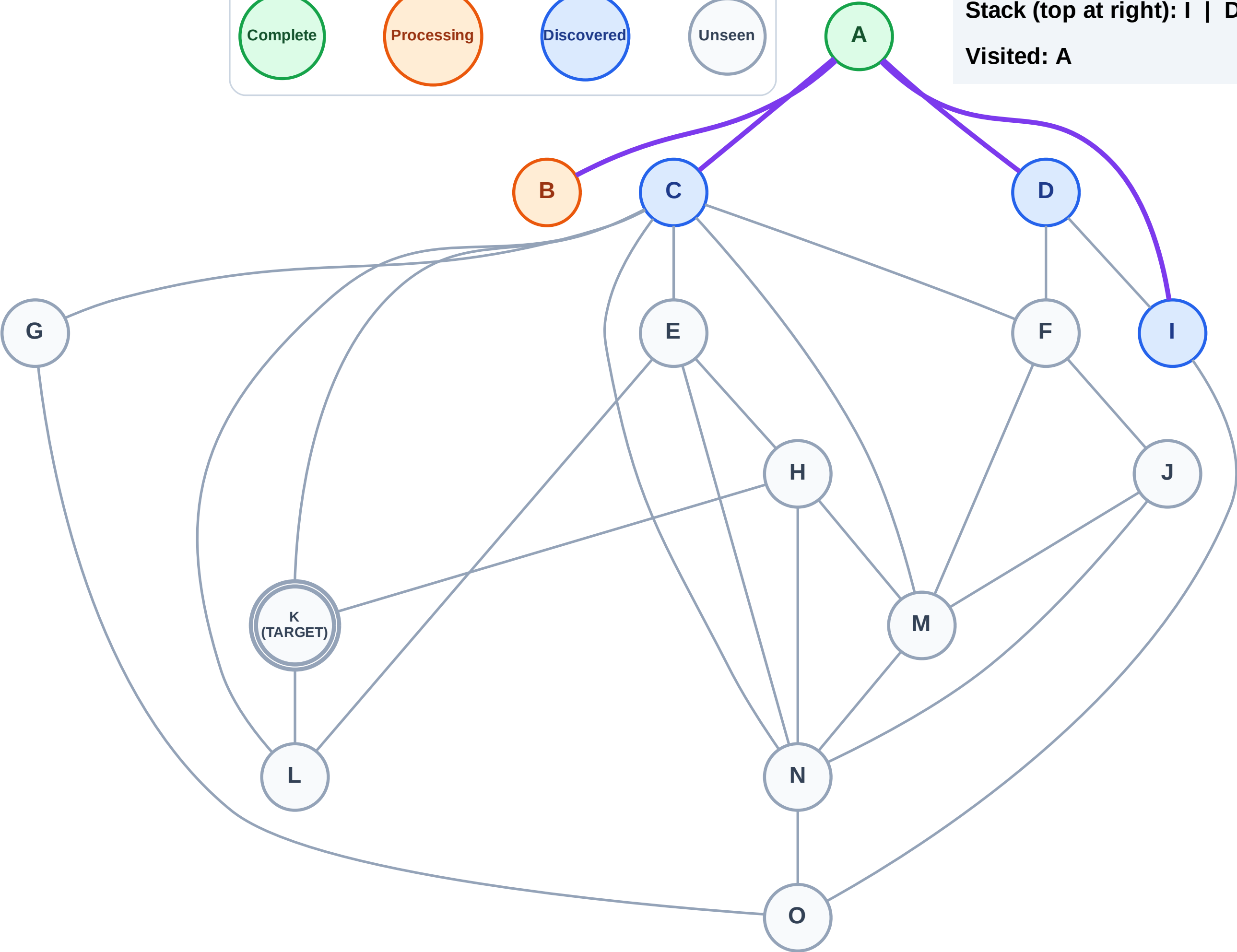
Processing

Discovered

Unseen

Stack (top at right): I | D | C

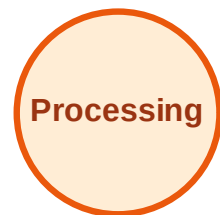
Visited: A



DFS Traversal

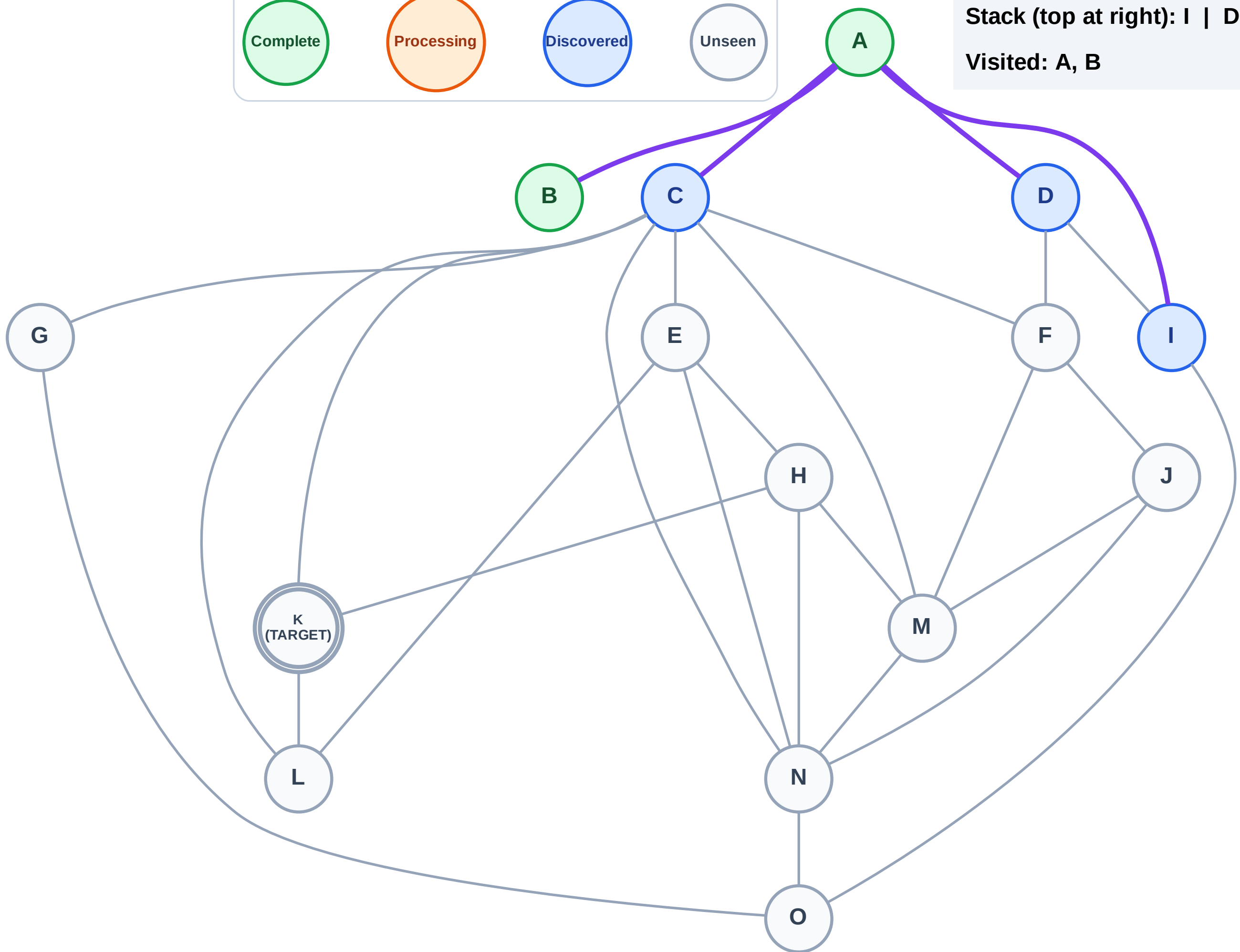
Step 5: No new neighbors from B

Legend



Stack (top at right): I | D | C

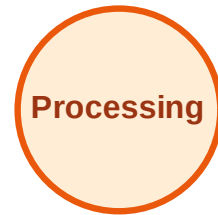
Visited: A, B



DFS Traversal

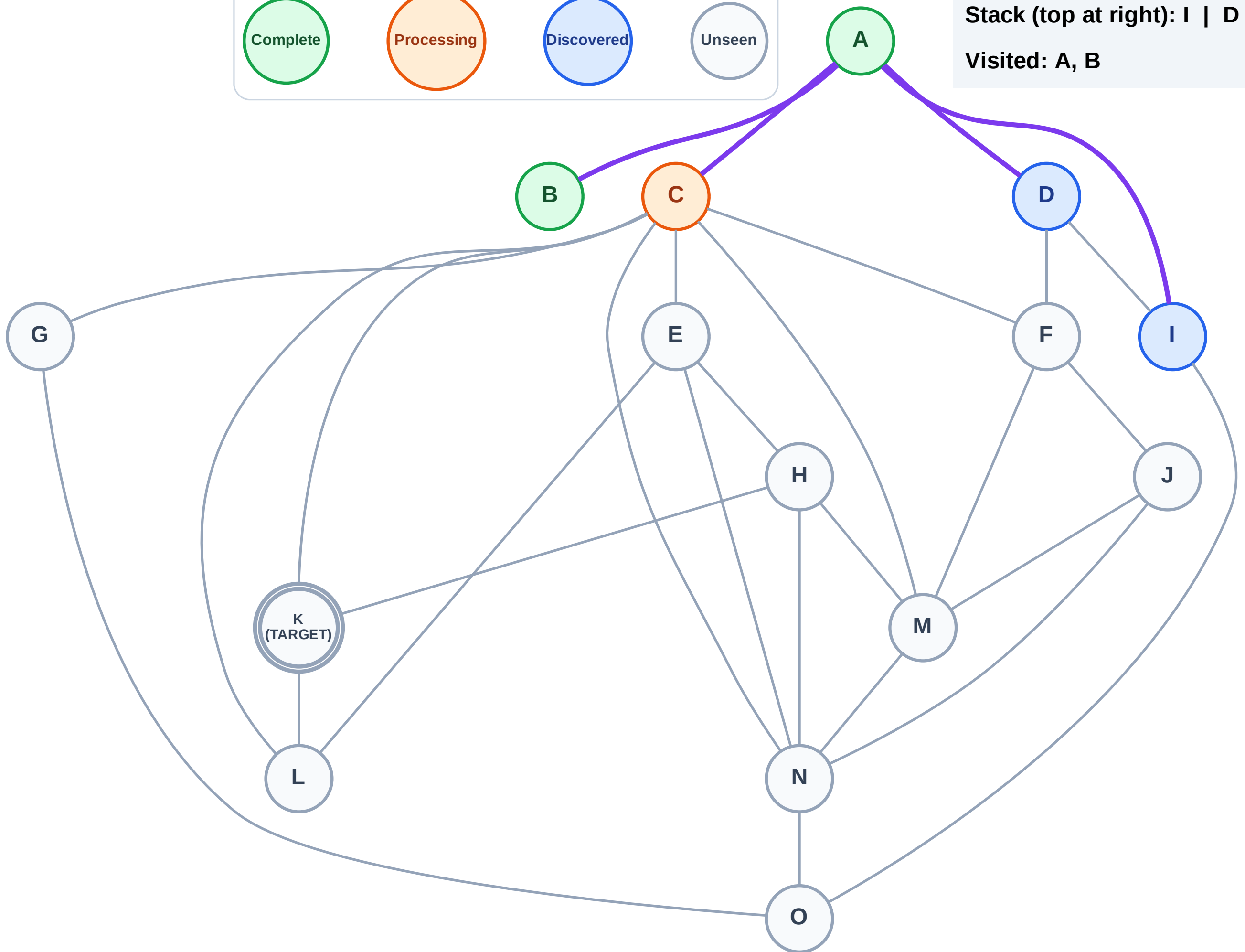
Step 6: Process node C

Legend



Stack (top at right): I | D

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): I | D | N | M | L
Visited: A, B, C, K

